Retrieval Augmented Generation (RAG)

Koustav Rudra
Assistant Professor
AI, IIT Kharagpur





What We'll Cover

- Motivating RAG
- Introduction to RAG
- Elaborate retrieval phase
- Common problems of RAG
- RAG variants
- RAG assessment framework


INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
1951
योगः कर्मसु कौशलम्

Why RAG?

Limitations with LLMs



LLMs can do a lot of tasks...

- Summarize Text
- Generate Code
- Rewrite Content

...but they suffer from

- Hallucinations
- Verifiability issues
- Knowledge cut-offs
- Learning failures

Limitations with LLMs: Hallucinations



What is the use of PMI of NLP?



In natural language processing (NLP), Pointwise Mutual Information (PMI) is often used as a measure of the association between words in a corpus. It can be used to identify words that are highly correlated and occur frequently together, which can be useful…



Can you give me citation for the above examples?



Here are some references for the examples of using PMI in natural language processing (NLP) that I provided:

Language modeling: “Pointwise Mutual Information (PMI) for Language Modeling” by R. B. Frazer and J. H. Lafferty (https://www.aclweb.org/anthology/W00-1308.pdf)
Information retrieval: “Information Retrieval using PMI-based Word Representations” by T. Mitamura and K. Kise (https://www.aclweb.org/anthology/C04-1100.pdf)
Machine translation: “PMI-based Word Alignment for Statistical Machine Translation” by K. Toutanova and C D. Manning (https://www.aclweb.org/anthology/W03-0419.pdf)

Limitations with LLMs: Hallucinations

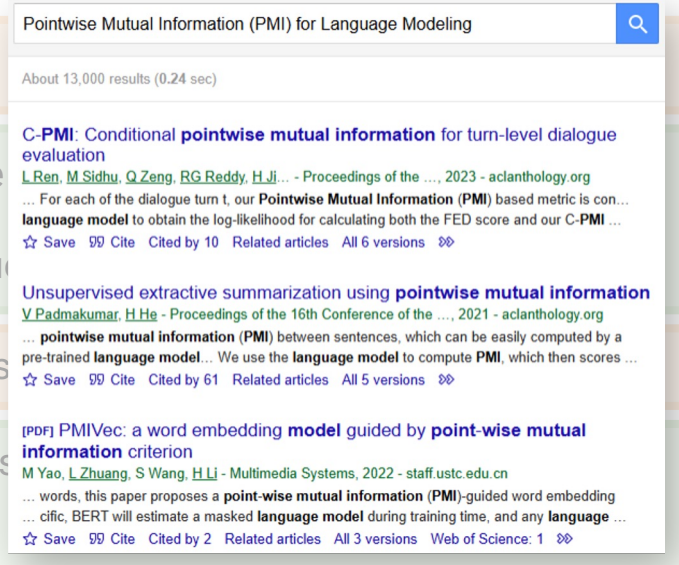
What is the use of PMI of NLP?

In natural language processing (NLP), Pointwise
as a measure of the association between words
words that are highly correlated and occur frequ

Can you give me citation for the above examples

Here are some references for the examples of us
(NLP) that I provided:

Language modeling: "Pointwise Mutual Information (PMI) for Language Modeling" by R. B. Frazer and J. H. Lafferty (https://www.aclweb.org/anthology/W00-1308.pdf)
Information retrieval: "Information Retrieval using PMI-based Word Representations" by T. Mitamura and K. Kise (https://www.aclweb.org/anthology/C04-1100.pdf)
Machine translation: "PMI-based Word Alignment for Statistical Machine Translation" by K. Toutanova and C D. Manning (https://www.aclweb.org/anthology/W03-0419.pdf)



Limitations with LLMs: Hallucinations

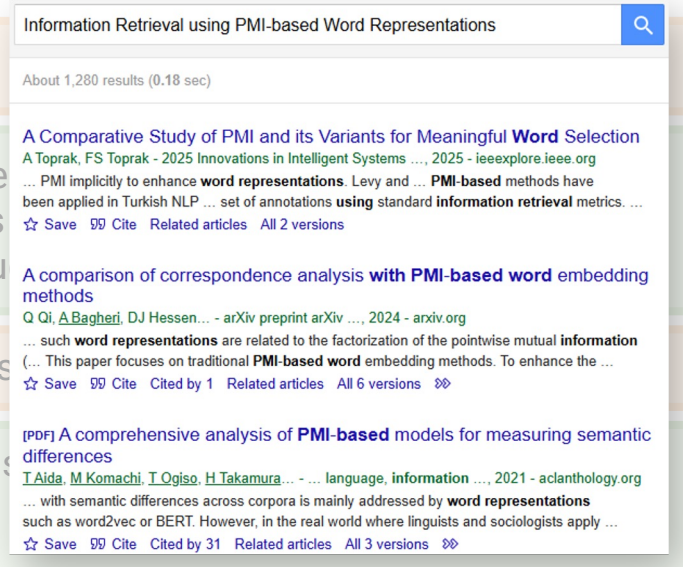
What is the use of PMI of NLP?

In natural language processing (NLP), Pointwise
as a measure of the association between words
words that are highly correlated and occur frequ

Can you give me citation for the above examples

Here are some references for the examples of us
(NLP) that I provided:

Language modeling: “Pointwise Mutual Information (PMI) for Language Modeling” by R. B. Frazer and J. H. Lafferty (https://www.aclweb.org/anthology/W00-1308.pdf)
Information retrieval: “Information Retrieval using PMI-based Word Representations” by T. Mitamura and K. Kise (https://www.aclweb.org/anthology/C04-1100.pdf)
Machine translation: “PMI-based Word Alignment for Statistical Machine Translation” by K. Toutanova and C D. Manning (https://www.aclweb.org/anthology/W03-0419.pdf)



Limitations with LLMs: Hallucinations



Dangers of hallucination:

- Recipe with incompatible ingredients
- Medical misguidance
- Misidentification of military targets
- Fabricated engineering designs with invalid parameters
- Altering or deleting production databases with confidence

Limitations with LLMs: Verifiability

What would have Richard Nixon said if the Apollo landing failed?

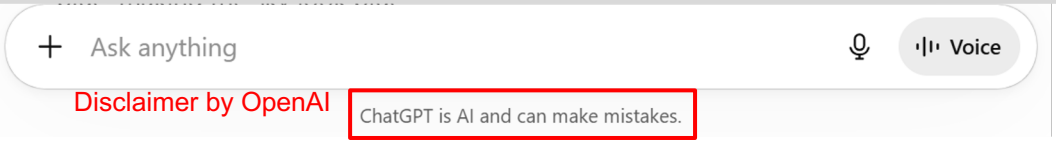
There's no historical record of a "backup" speech for a failed Apollo 11 landing, so anything here is necessarily speculative—but we can infer what a statement from Richard Nixon would likely have emphasized based on how NASA and the White House prepared for worst-case scenarios.

...

A plausible version of his remarks might have been:

> "Today, we have suffered a tragedy in our quest to reach the Moon. The astronauts of Apollo 11 set out in the noblest tradition of exploration, and they did so on behalf of all mankind.
>
> Their courage and sacrifice will not be forgotten. In them, we see the very best of human determination and spirit.
>
> Though this mission has ended in sorrow, the United States will continue to pursue space exploration, not in recklessness, but in honor of those who dared greatly."

...



Disclaimer by OpenAI

Limitations with LLMs: Verifiability



Dangers with verifiability:

- Distorted historical narratives (historians believe that…)
- Fabricated expertise (according to my analysis…)
- Personal manipulation (based on your habits…)
- Religious or spiritual claims
- Political manipulation

Limitations with LLMs: Knowledge-cutoff

Who is the winner of IPL 2026?

I don't know who won the IPL 2026 title.

My knowledge doesn't include verified real-world results from events that may have occurred after my training data, and I don't have live access to sports results in this conversation.

If the IPL 2026 season has already concluded, you can check the official IPL website or a sports news source, and I can help analyze the tournament, teams, or statistics if you share the result.

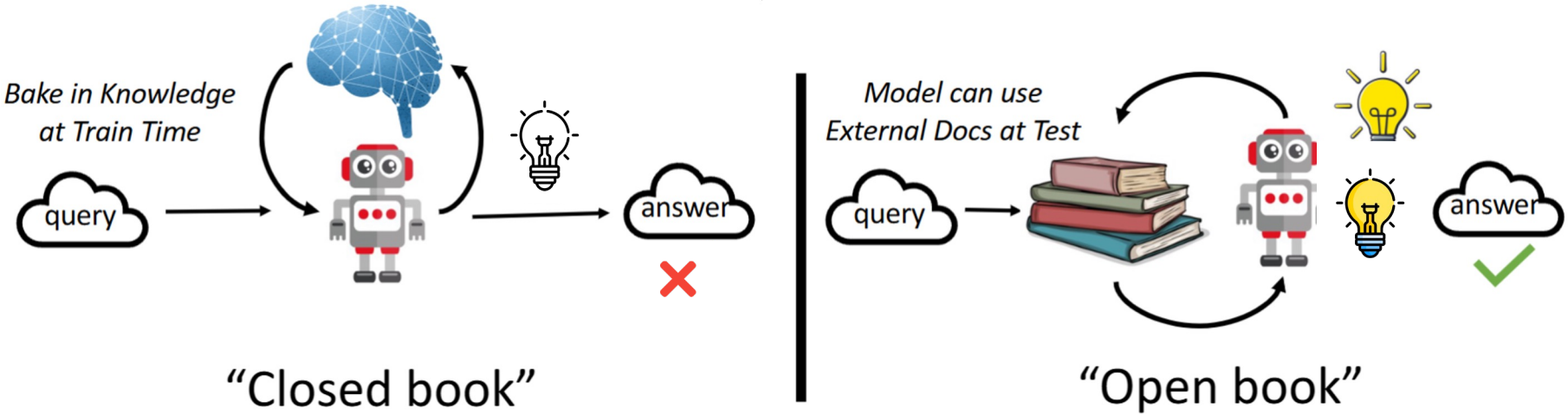
Limitations with LLMs: Knowledge-cutoff



Dangers with knowledge-cutoff:

- Outdated medical information
- Security vulnerabilities
- Stale confidence (The recommended procedure is…)

Motivation for RAG



Motivation for RAG

What LLMs don't know (the open book)

Private Databases
LLMs can't access confidential information

Hard to access information
Some information isn't widely available online, making it inaccessible to LLMs

Real time data
LLMs are trained on past data and don't update automatically

How to make sure the LLM knows this useful information?

Motivation for RAG



Put it in the prompt

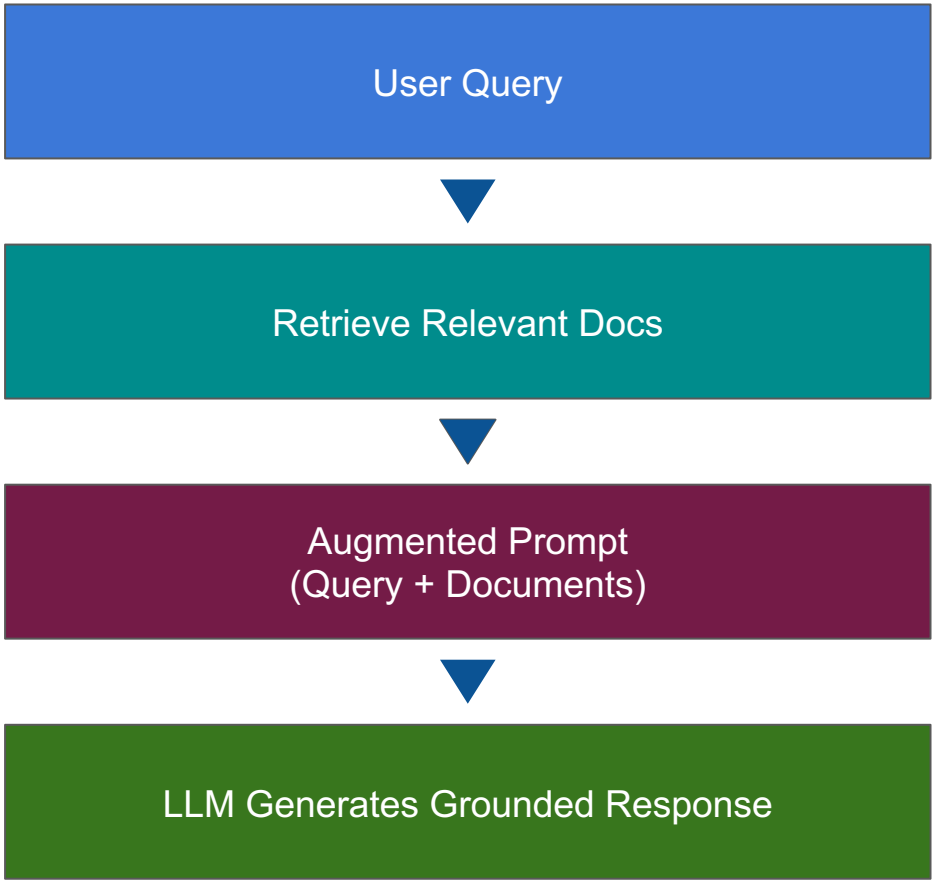




Introduction to RAG

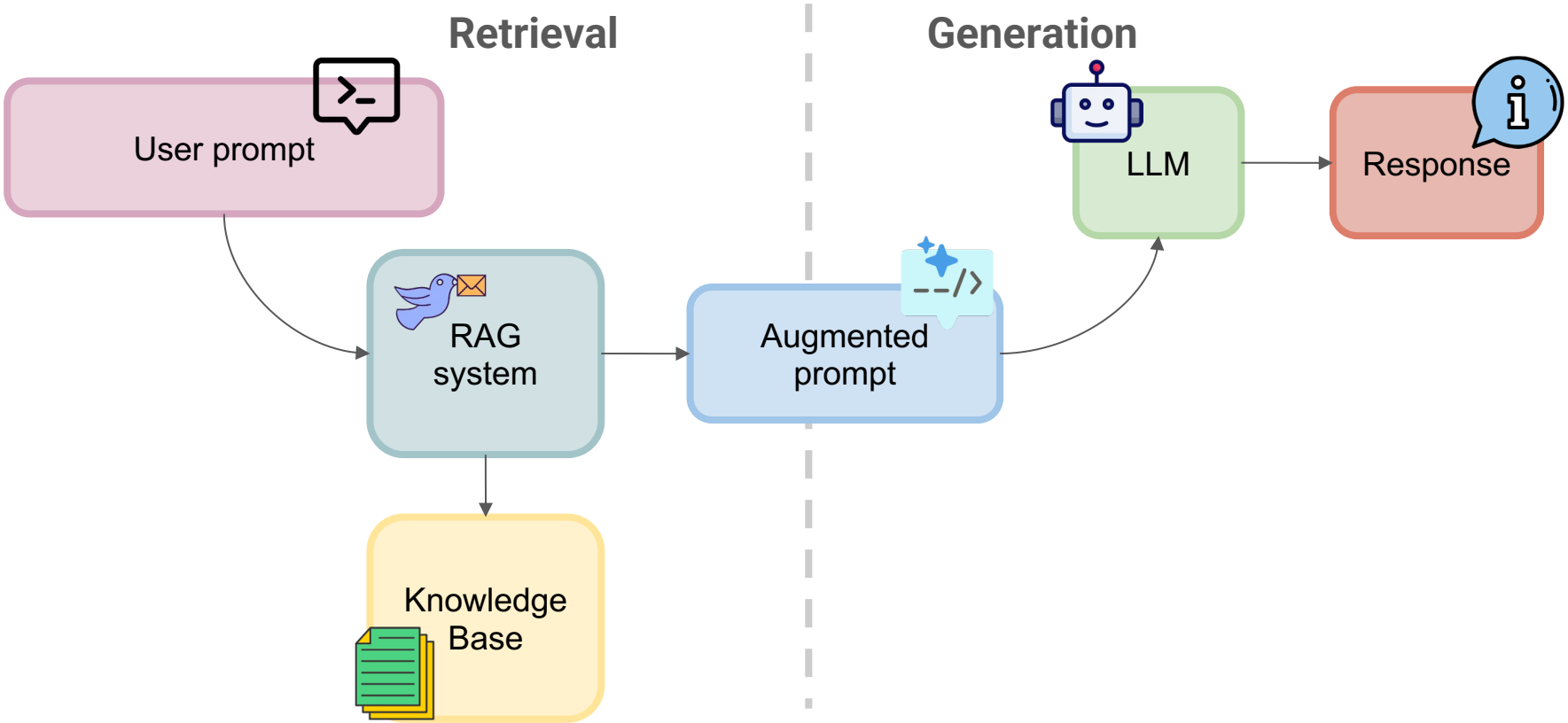
Overview of RAG

The RAG Solution

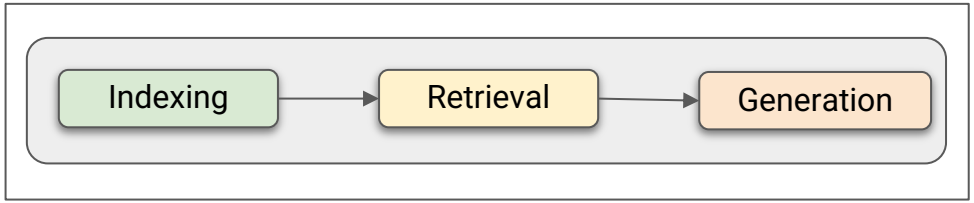


Lewis et. al. (2020). NeurIPS

RAG architecture



RAG Workflow



Basic RAG Workflow

Motivation: LLMs can benefit from access to external knowledge sources for factual accuracy and reliability.

How (Basic) RAG Works:

- Input: User provides a query.
- Retrieval: Relevant documents are retrieved from a knowledge source (e.g., Wikipedia).
- Generation: The LLM uses the retrieved information to generate a response.

Benefits:

Factual Consistency: Improves the accuracy of generated content.

Up-to-Date Information: Allows LLMs to access current information beyond their training data.

RAG Workflow

Problem: LLMs can struggle with factual accuracy and limited pretrained information

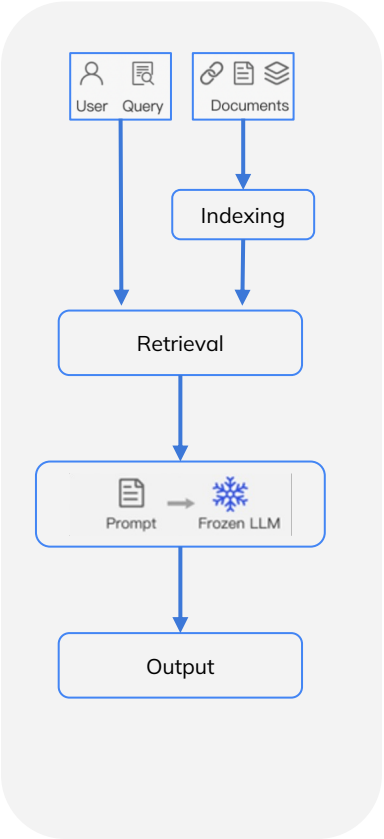
Solution: Give LLMs access to external knowledge sources.

RAG Approach:

- User asks a question.
- System finds relevant documents
- LLM uses these documents to generate an accurate and reliable answer.

Benefits:

- More factual responses.
- Access to up-to-date information.



Naive RAG

RAG Workflow

Advanced RAG: On top of the naive RAG workflow, we can add a pre-retrieval and a post retrieval step

Pre-retrieval: rewrites user query

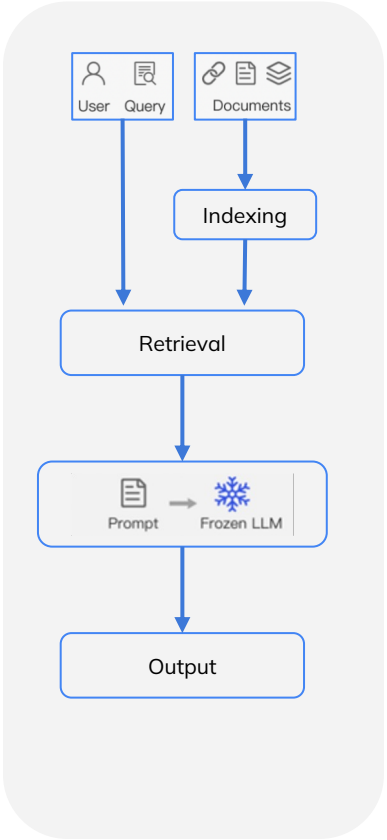
Benefits:

- ➢ Match domain-specific terminologies
- ➢ Provide additional context

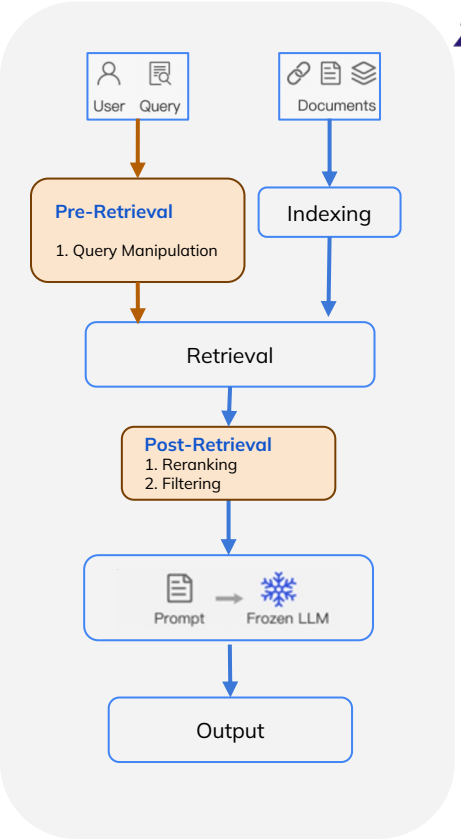
Post-retrieval: improves quality of added documents

Benefits:

- ➢ Reranks retrieved documents to reject irrelevant ones
- ➢ Filter out low-value contents



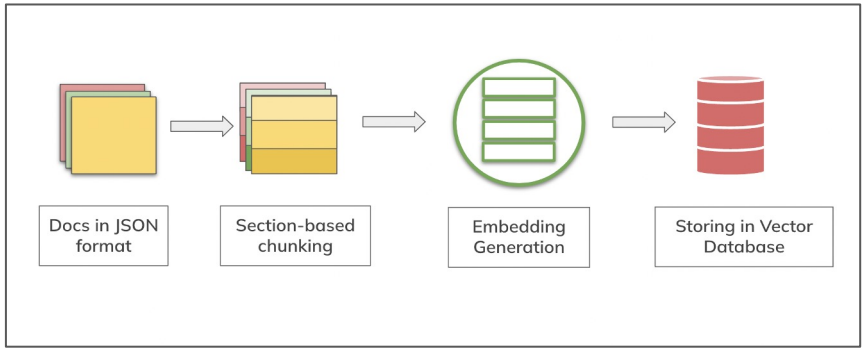
Naive RAG



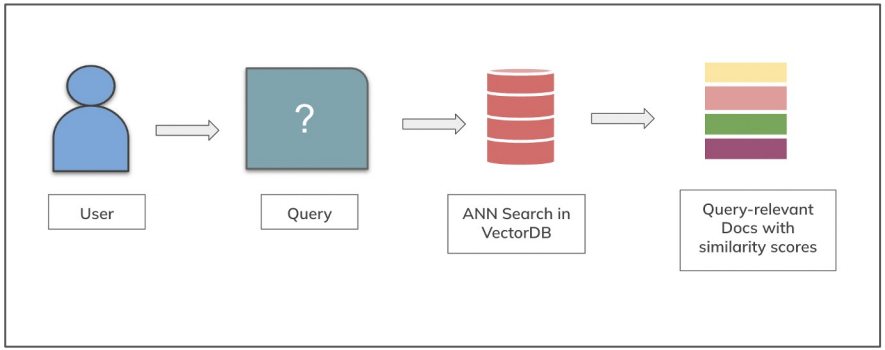
Advanced RAG

RAG: Example

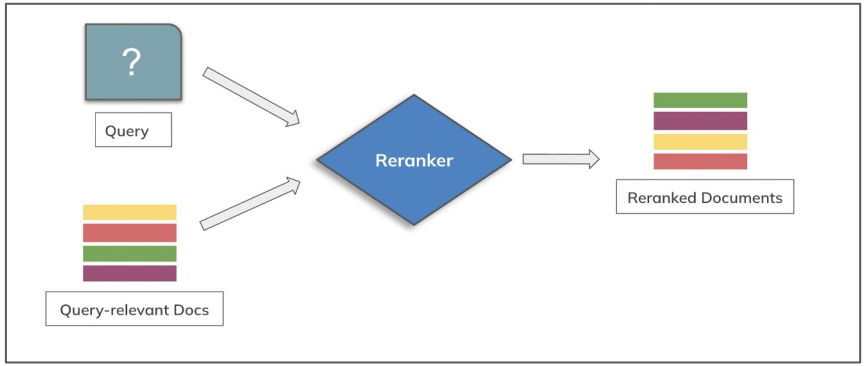
Pre-Retrieval



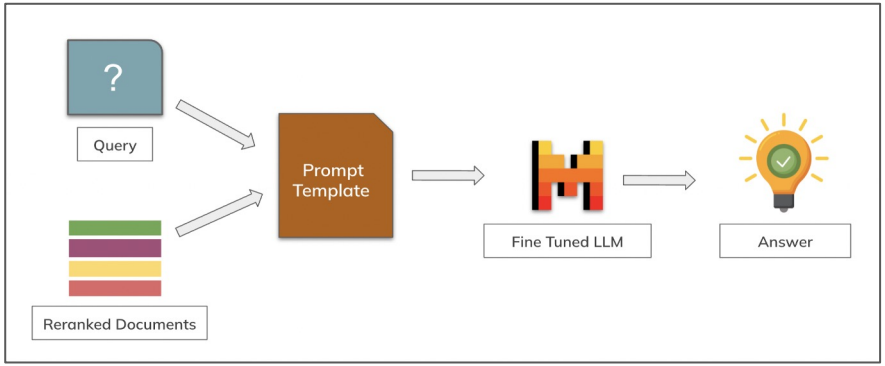
Retrieval



Post-Retrieval



Augmentation and Generation



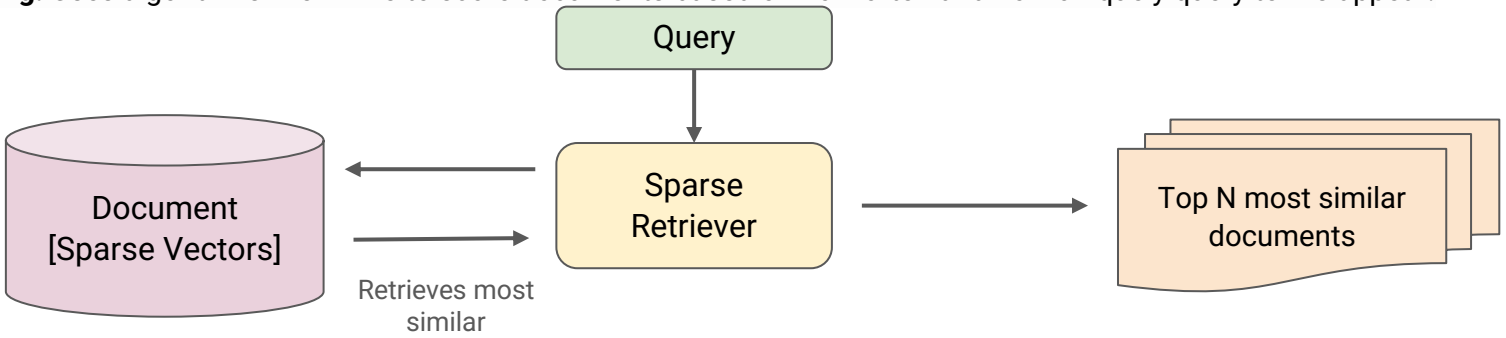
https://ai.meta.com/blog/retrieval-augmented-generation-streamlining-the-creation-of-intelligent-natural-language-processing-models/

Sparse Retrieval

Sparse Retrieval in RAG: Keyword-Focused Search

Method: Matches documents based on shared keywords with the user's query.

Ranking: Uses algorithms like BM25 to score documents based on how often and how uniquely query terms appear.



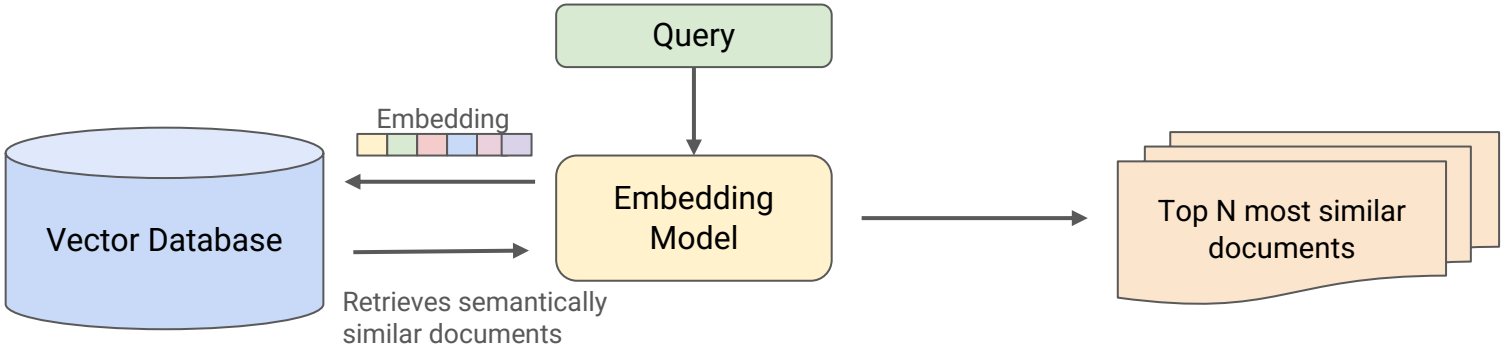
Strengths: Fast and efficient, especially for large datasets.

Weakness: Limited understanding of meaning; might miss relevant documents that use different words.

- Sparse retrieval methods are often more interpretable, as they provide clear reasoning based on term occurrences, making it easier to understand why certain documents were retrieved.

Dense Retrieval

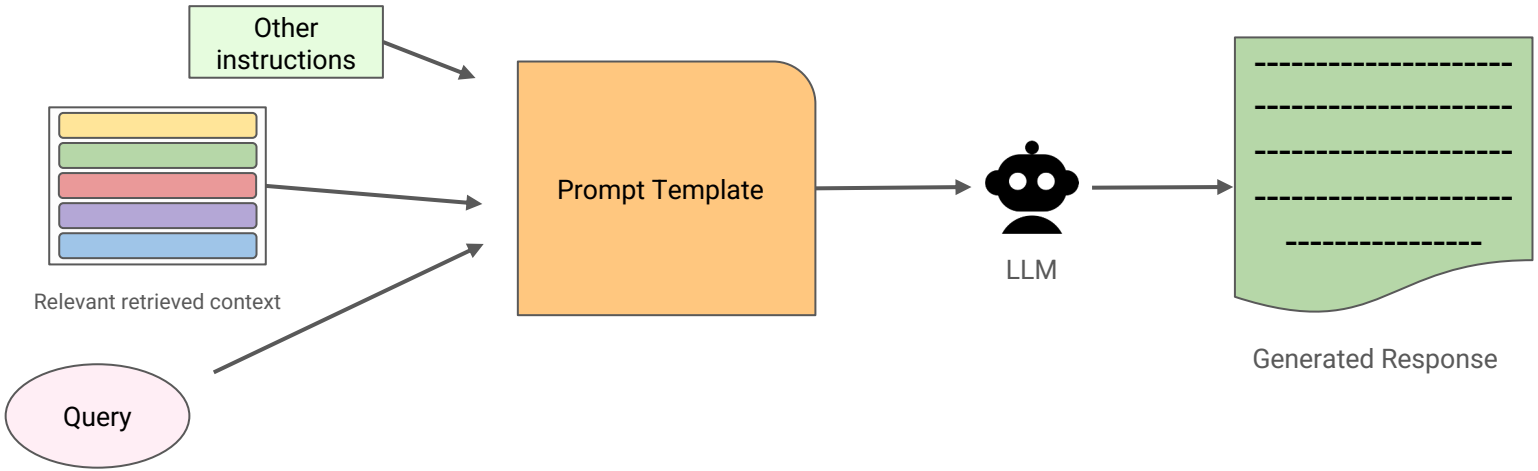
- **Method:** Uses deep learning (e.g., BERT, T5) to represent both queries and documents as vectors (embeddings) that capture semantic meaning.
- **Matching:** Finds documents with similar meanings to the query, even if they don't share exact keywords.



- **Strengths:** More accurate and relevant retrieval, especially for complex or nuanced queries.
- **Weaknesses:** Can be computationally more expensive than sparse methods.

Augmentation and Generation

- **Grounding :** Grounding involves providing LLMs with use-case specific data and knowledge sources that are not inherently part of their original training data. RAG framework acts as grounding for LLMs



- The LLM processes the input and generates a coherent and contextually appropriate response.
- This response leverages the external information retrieved in the augmentation step, allowing the model to provide more accurate and nuanced answers.

Elaborate retrieval phase

Retrieval pipeline

Raw documents are converted to retrieved chunks

1

Document Ingestion

Load PDFs, HTML, databases. Handle multiple formats.

2

Chunking

Split docs into overlapping segments (e.g. 256-512 tokens).

3

Encoding / Indexing

Embed chunks (dense) or build inverted index (sparse).

4

Query Processing

Encode the user query into same representation as the index

5

Retrieval & Ranking

Score candidate chunks (cosine sim / BM25). Return top-k most relevant.

6

Re-ranking (optional)

Cross-encoder re-ranks top-k results. Improves precision.

Chunking and Query Formulation

Chunking Strategies

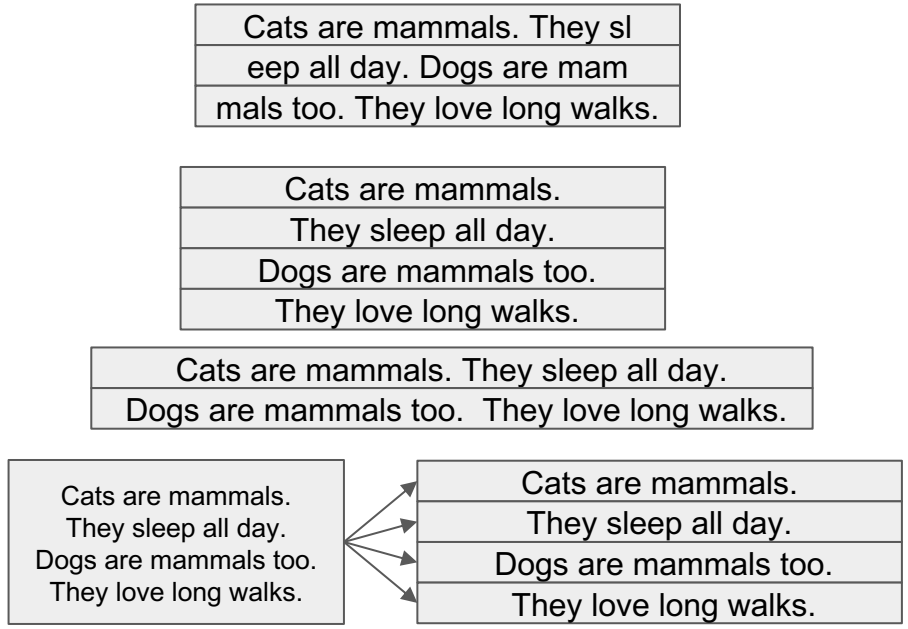
Fixed-size chunking
Simple sliding window (e.g. 512 tokens, 10% overlap). Fast but may split mid-sentence.

Sentence / Paragraph
Split of natural sentence boundaries. Better coherence; relies on good sentence detection.

Semantic chunking
Group semantically related sentences together using embedding similarity shifts.

Hierarchical / parent-child
Store small chunks for retrieval, retrieve larger chunk to send to LLM (better context).

Example: "Cats are mammals. They sleep all day. Dogs are mammals too. They love long walks."



Chunking and Query Formulation

Query Formulation

Naive: query as-is
Send raw user question directly to retriever. Works for simple, well-formed queries.

HyDE (Hypothetical Document)
Ask LLM to generate a hypothetical answer first; use that as the retrieval query.

Multi-query
Generate multiple paraphrased queries; union the results. Handles ambiguity.

Step-back prompting
Abstract the query to a higher-level concept before retrieval. Improves coverage for complex questions.

Example: "How do I reduce my electricity bill?"

How do I reduce my electricity bill?

Lower your electricity bill by switching to LED bulbs, unplugging idle devices, using energy-efficient appliances, and adjusting your thermostat.

Ways to lower home energy costs

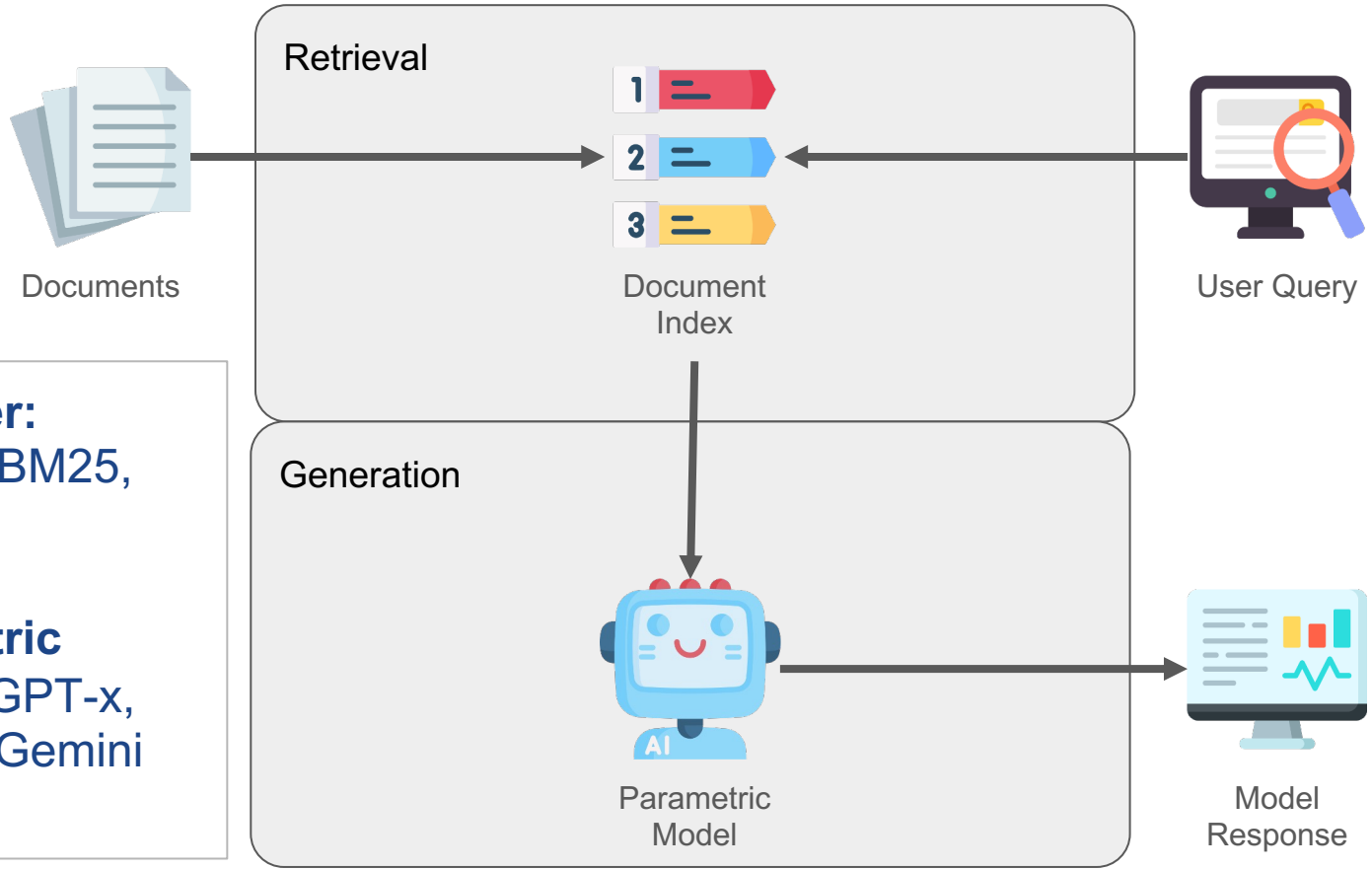
Tips to save on power consumption

How to make appliances more energy efficient

What factors affect household electricity usage?

[2212.10496] Precise Zero-Shot Dense Retrieval without Relevance Labels
[2305.14283] Query Rewriting for Retrieval-Augmented Large Language Models

Retriever



Retriever: Google, BM25, Elser

Parametric model: GPT-x, Claude, Gemini etc.

Information retrieval methods

TF-IDF

tf-idf (term frequency-inverse document frequency)



Measure of importance of words in a document

- term frequency
  - How frequent a term appears in a document
  - More frequent words are aligned with the topic of the document

- inverse document frequency
  - Inversely proportional to the number of documents containing the term
  - Terms present in many documents are not significant



TF-IDF

E.g.: Consider the following three documents:



d1

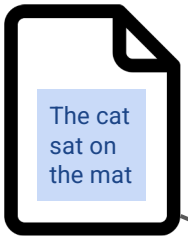


d2



d3

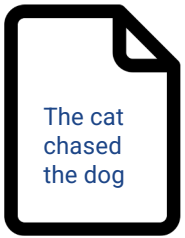
TF-IDF



d1



d2



d3

N = number of documents
d = current document
t = current term
df(t) = how many document contain t

$$tf(t,d) = count(t \text{ in } d) / \text{total terms in } d$$
$$idf(t) = \log(N / df(t))$$
$$tf\text{-}idf(t,d) = tf(t,d) * idf(t)$$

Term	TF	df	IDF	TF-IDF
the	0.333	3	0.000	0.000
cat	0.167	2	0.176	0.029
sat	0.167	2	0.176	0.029
on	0.167	2	0.176	0.029
mat	0.167	1	0.477	0.080

BM25: A Nonbinary Model

The simplest score for document d is just idf weighting of the query terms present in the document:

Retrieval Status Value (RSV): $$RSV_d = \sum_{t \in q} \log \frac{N}{\mathrm{df}_t}$$

Improve this formula by factoring in the term frequency and document length:

$$RSV_d = \sum_{t \in q} \log \left[\frac{N}{\mathrm{df}_t} \right] \cdot \frac{(k_1 + 1)\mathrm{tf}_{td}}{k_1((1 - b) + b \times (L_d / L_{\mathrm{ave}})) + \mathrm{tf}_{td}}$$

- tf_{td} : term frequency in document d
- L_d (Lave): length of document d (average document length in the whole collection)
- k_1: tuning parameter controlling the document term frequency scaling
- b: tuning parameter controlling the scaling by document length

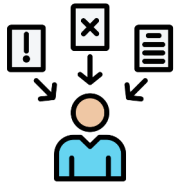
Common problems of RAG

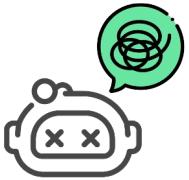
RAG failure modes





- Retrieval failures
- Context Failures
- Generation Failures







RAG failure modes: Retrieval failures

Example Query: What is the return window for online orders?
Relevant chunk: Online orders can be returned for a refund within 30 days of delivery.

Retrieved chunk:

Wrong chunk retrieved	“Our in-store exchange policy allows swaps within 14 days” Wrong answer: 14 days.
Semantic similarity neq relevance (vocabulary mismatch)	
Incomplete retrieval	“Online orders can be returned for a refund” Wrong answer: I don’t know.
Multi-hop questions need facts from multiple docs	
Stale knowledge base	“Online return window: 15 days.” (last year’s policy) Wrong answer: 15 days.
Docs not updated, retriever returns outdated information	

RAG failure modes: Context Failures

Example Query: When was the company founded?

Retrieved chunk(s):

Lost in the middle LLMs ignore chunks in the middle of long contexts	[chunk 1] Office locations… [chunk 2] Mission statement… [chunk 5] Founded in 2008 < correct chunk [chunk 9] Product list… [chunk 10] Contact info… Answer: I don't know
Context overload Too many retrieved chunks degrade generation quality	[5 chunks about history, HR policies, blog posts, faqs…] Answer: vague / mixed-up
Irrelevant context Noisy chunks mislead the LLM even when good chunks are also present	"Our partner company was founded in 1999." Wrong answer: 1999.

[2307.03172] Lost in the Middle: How Language Models Use Long Contexts
[2302.00093] Large Language Models Can Be Easily Distracted by Irrelevant Context

RAG failure modes: Generation Failures

Example Query: What is the refund window for online orders?

Retrieved chunk(s):

Hallucination despite retrieval LLM still generates facts not in retrieved docs	"Refundable within 30 days" Answer: You get 30 days, and a free shipping label plus a 10% voucher.
Knowledge conflict Retrieved doc contradicts model's parametric memory	Context: "30 days" Parametric memory: "most stores use 15 days" Answer: 15 days
Attribution errors Model cites wrong source or ignores the provided context	"Refundable within 30 days" [Policy.pdf] Answer: 30 days. [Source: Shipping-FAQ.pdf]

[2311.05232] A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions

Types of RAG

- *Standard RAG*
- *RAG with memory*
- *Agentic RAG*
- *Chain-of-RAG*

Standard RAG

It is **Stateless:** every query is treated independently

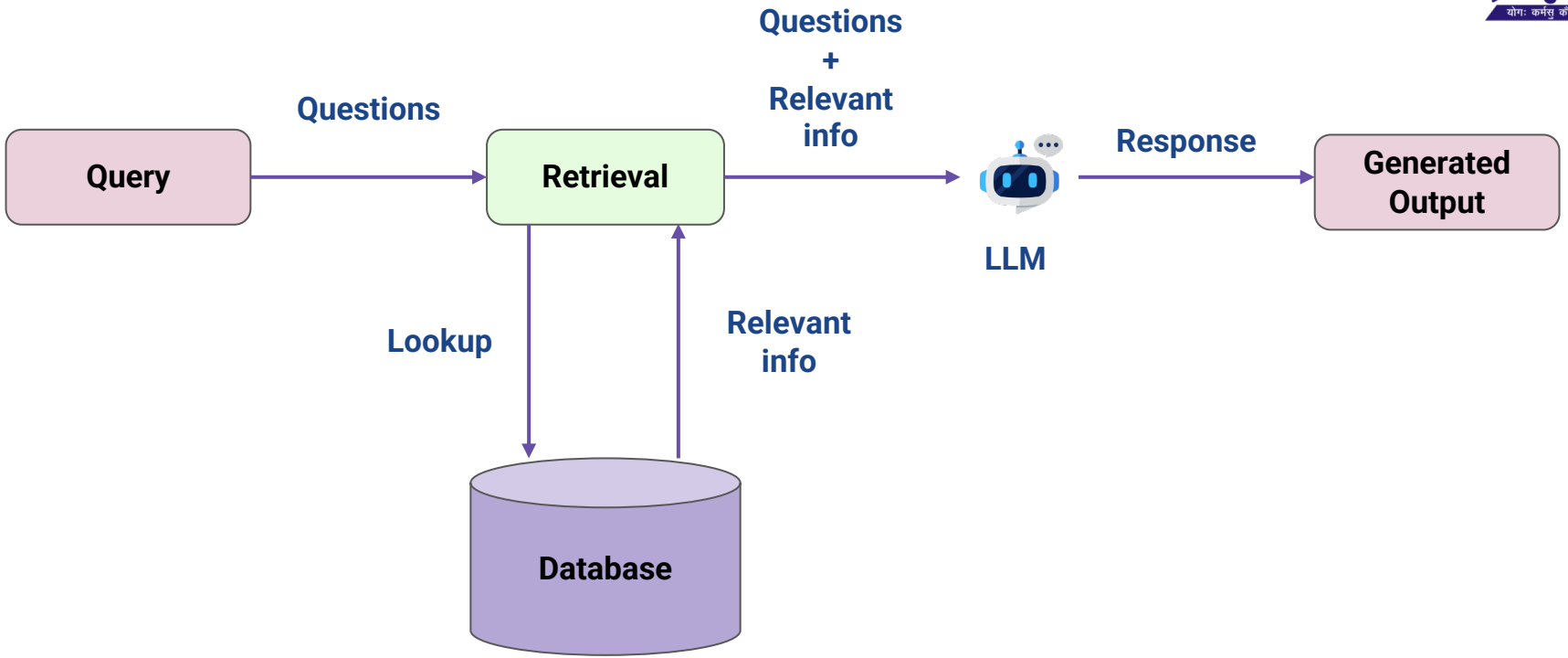
Example:

Q: "What is the boiling point of water at sea level?"

- **system embeds the query**
- **looks up the closest passage in the document store (e.g., a physics reference)**
- **LLM answers "100°C" grounded in that retrieved passage.**

https://arxiv.org/abs/2005.11401

Standard RAG



[2005.11401] Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Standard RAG with memory

User asks:

- *Q1: "I'm a vegetarian."* 🌱
- *Q2: "Suggest high-protein foods."*

Some use-cases:

- personal AI agents
- conversational chatbots
- customer support systems
- educational tutoring platforms

- It adds a memory of past turns.
- Memory is the system's ability to keep track of past interactions such as past questions, answers, or retrieved documents.
- It doesn't just remember what was said, but understands how previous context can influence new searches.

Standard RAG with memory

Example:

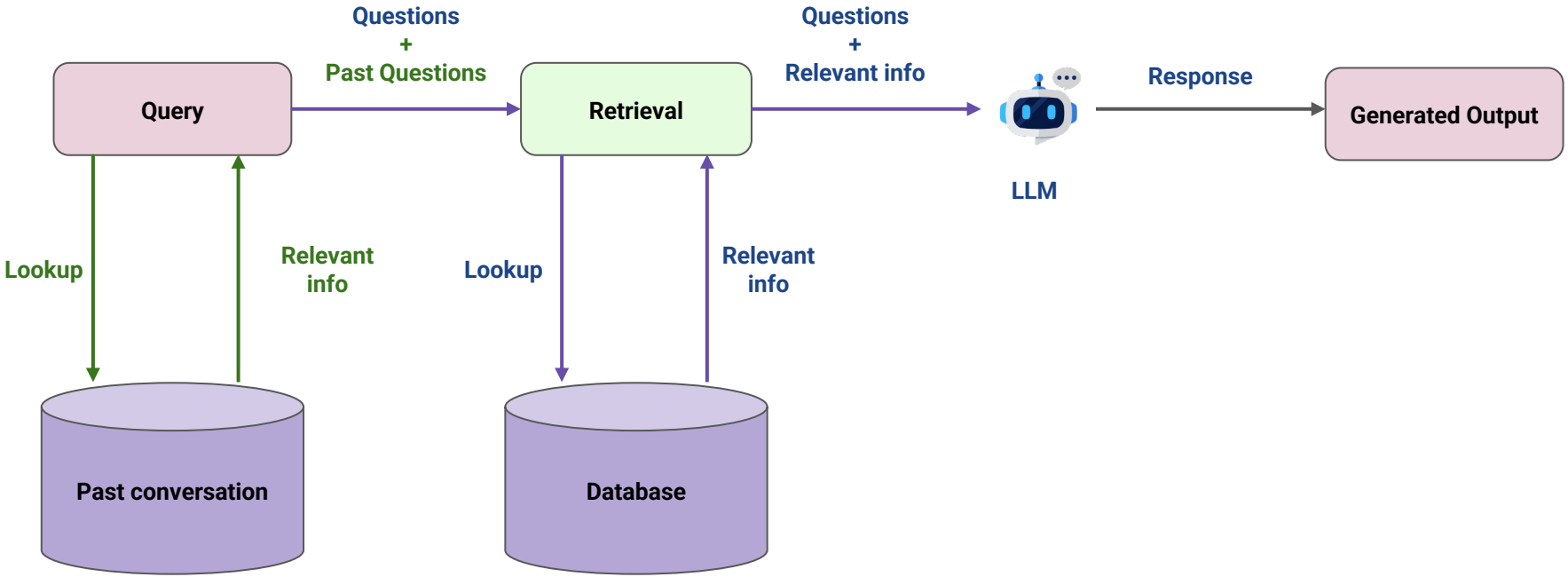
Q1: "What is the capital of France?"

A1: "Paris."

Q2: "What about its population?"

- **Here, user asks about the capital of France and later refers to 'its population,'**
- **system recalls the context to determine the query is still referring to Paris.**
- **rewrites the query to "What is the population of Paris?"**
- **Final answer: Paris has about 2.05 million**

Standard RAG with memory



https://arxiv.org/abs/2005.11401

Agentic RAG

Example:

Q1: "How many more people live in Paris's metropolitan area than in the city proper?"

- **Steps (the agent plans and picks tools):**
  - Plan: decompose the two part, needs two figures, then do subtraction.
  - Web Search / collection lookup: city proper ≈ 2.1M; metro ≈ 11.3M. The current metro area population of Paris in 2025 is 11,347,000.
  - Reflect: "Do I have both numbers? Yes." → proceed.
  - Calculator tool: 11.3M − 2.1M = 9.2M.

Final answer:

“About 9.2 million more people live in the Paris metropolitan area than in the city proper.”

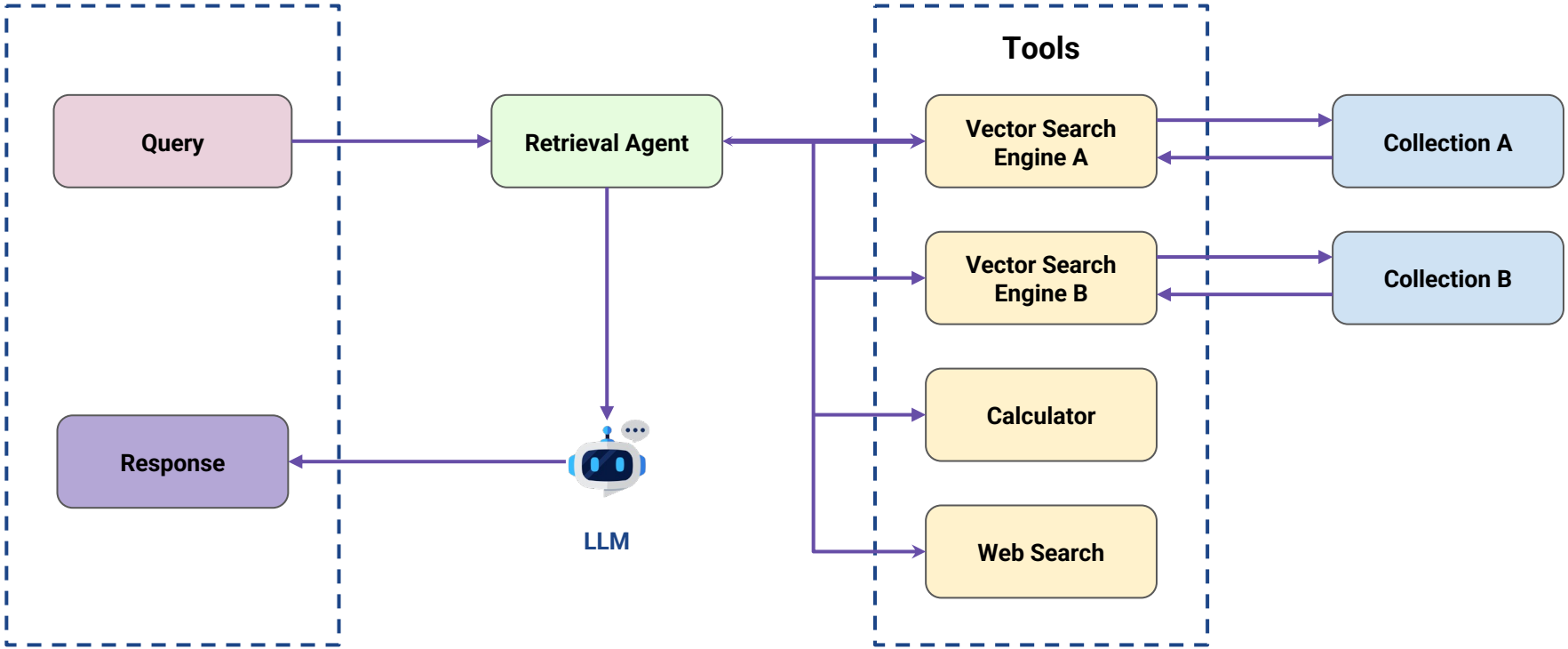
Agentic RAG

- Instead of just retrieving the first relevant documents, it plans its approach
- decides what to investigate
- takes action using tools.
- it breaks a task into smaller steps and searches various data sources
- checks whether what it found answers the question, and if not, it keeps searching.

Some use-cases:

- legal research where lawyers conduct comprehensive case analysis
- financial analysis that combines market data with regulatory information

Agentic RAG


Query
Retrieval Agent
Tools
Vector Search Engine A
Collection A
Vector Search Engine B
Collection B
Calculator
Web Search
Response
LLM

Chain-of-RAG (CoRAG)

What if my question looks like this?

"What's the nationality of the director of the film that won Best Picture the year Titanic was released?"

The Problem with Traditional RAG:

- **Imperfect Retrieval:** a single-step retrieval often fails to gather all relevant information for complex tasks
- **Static Reasoning:** the lack of iterative refinement leads to suboptimal generation
- **Contextual Misalignment:** queries are not adapted dynamically as new information surfaces

Chain-of-RAG (CoRAG)

What CoRAG Does Differently?

- CoRAG extends the RAG framework by enabling dynamic, multi-step retrieval.
- Rather than stopping after one lookup, it iteratively refines queries based on intermediate states, mirroring how humans approach problem-solving.
- This makes it especially strong at multi-hop question answering.

- **Sub-query 1:** "When was Titanic released?" → 1997.
- **Sub-query 2:** "What won Best Picture for 1997 films?" → Titanic itself (at the 1998 ceremony).
- **Sub-query 3:** "Who directed Titanic?" → James Cameron.
- **Sub-query 4:** "What is James Cameron's nationality?" → Canadian.

CoRAG: Mechanics

How it works?

- **Dynamic Retrieval Chains:** starting from the initial query, CoRAG iteratively refines and retrieves sub-queries and sub-answers until sufficient information is gathered
- **Rejection Sampling for Training:** it generates intermediate retrieval chains to augment training datasets, improving the model's reasoning capabilities
- **Flexible Decoding Strategies:** at inference it uses methods like greedy decoding, best-of-N sampling, and tree search to balance performance and computational cost

CoRAG: More examples

Example 2:

Question: *What experiments led to the discovery of gravitational waves?*

Step 1: Sub-query: "What are gravitational waves?"
Sub-answer: Ripples in spacetime caused by massive accelerating objects

Step 2: Sub-query: "Which experiment first detected gravitational waves?"
Sub-answer: The LIGO experiment in 2015

Final Answer: The LIGO experiment in 2015 detected gravitational waves, confirming Einstein's theory of general relativity

Example 3:

Question: *What is the nationality of the director of the 1994 film that won Best Picture?*

Step 1: Sub-query: "Which film won Best Picture in 1994?"
Sub-answer: Forrest Gump

Step 2: Sub-query: "Who directed Forrest Gump?"
Sub-answer: Robert Zemeckis

Step 3: Sub-query: "What is Robert Zemeckis's nationality?"
Sub-answer: American

Final answer: American

RAG: Assessment Framework

RAG boosts LLMs with external knowledge, but introduces failure points the source article identifies:

- **Retrieval issues** — wrong documents retrieved or key information missing
- **Hallucination risk** — the LLM generates content not in the retrieved documents
- **Context utilization** — the LLM fails to use retrieved information effectively

Without evaluation, these lead to inaccurate or fabricated responses which is critical in high-stakes domains like pharmaceutical research where accuracy is paramount.

Two Leading Frameworks

- **RAGAS**
- **TruLens**

RAGAS

Retrieval-Augmented Generation Assessment Suite (RAGAS)

- It is an open-source, RAG evaluation framework
- It measures how well a system retrieves and utilises external context to generate accurate, faithful and relevant responses.
- RAGAS evaluates the entire RAG pipeline i.e from context retrieval to answer generation

Answer Relevancy: Measures how closely the generated response matches the user's query intent.

Faithfulness: Evaluates whether the response stays true to the retrieved evidence.

Context Precision: Assesses how relevant the retrieved chunks are to the query.

Context Recall: Determines whether all necessary information was retrieved.

Evaluation Dataset: Represents the structured input including questions, contexts, responses and reference answers used for metric computation

RAGAS

Example:

Field	Value
question	"Who wrote Romeo and Juliet?"
contexts (what the retriever pulled)	"Romeo and Juliet is a tragedy written by William Shakespeare."
answer (what the LLM generated)	"Romeo and Juliet was written by William Shakespeare in 1597."
ground_truth (optional reference)	"William Shakespeare wrote Romeo and Juliet."

Notice: the answer adds "in 1597", which is not in the retrieved context.

That's a hallucinated detail (even if it happens to be plausible).

Watch how only one metric catches it.

RAGAS

Example:

Metric	Score	Why
Context Precision	~1.0	The one retrieved chunk is fully on-topic for the question.
Context Recall	~1.0	The context contains everything the ground truth needs (the author).
Answer Relevancy	~0.95	The answer directly addresses "who wrote it."
Faithfulness	~0.5	Two claims in the answer: "written by Shakespeare" (supported by context) and "in 1597" (not in context). Half the claims are grounded, so the score drops.

Thank You

Instructor



Koustav Rudra
krudra@ai.iitkgp.ac.in

Teaching Assistants (IITKGP)



Anurag Sharma



Subhojyoti Khastagir